

Midterm 2 Review

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Exercise 1. We call an operator **normal** if it commutes with its hermitian conjugate, $[A, A^\dagger] = 0$. In particular, hermitian, anti-hermitian, and unitary operators are all normal, so this is a more general class of operators than hermitian operators.

- a) Show that if A is normal and $A|u\rangle = a|u\rangle$ for some nonzero $|u\rangle$, then $A^\dagger|u\rangle = a^*|u\rangle$.
 - b) Show that (a) implies that the eigenbras of A are the hermitian conjugates of the eigenkets, and the left spectrum is identical to the right spectrum.
 - c) Show that the eigenspaces corresponding to distinct eigenvalues of a normal operator are orthogonal. This is a generalization of the proof for hermitian operators.
- a) We want to show that

$$(A^\dagger - a^*)|u\rangle = 0.$$

Indeed, we can compute that

$$\langle u|(A - a)(A^\dagger - a^*)|u\rangle = \langle u|AA^\dagger|u\rangle + |a|^2\langle u|u\rangle - a\langle u|A^\dagger|u\rangle - a^*\langle u|A|u\rangle = 0,$$

by commutativity of A and $A^\dagger$.

- b) Let's take an eigenbra of A , say $\langle A\alpha| = \alpha\langle\alpha|$. Then $(\langle A\alpha|)^\dagger = A^\dagger|\alpha\rangle = \alpha^*|\alpha\rangle$, so we see that $|\alpha\rangle$ is an eigenket of $A^\dagger$ with eigenvalue α^* . But by (a), that means that it's also an eigenket of A with eigenvalue α . This also implies the second statement of (b).
- c) Suppose $|u\rangle, |v\rangle$ are eigenvectors in eigenspaces with distinct eigenvalues, say $A|u\rangle = a|u\rangle$ and $A|v\rangle = b|v\rangle$. We want to show that $\langle u|v\rangle = 0$. We see that

$$\langle u|A|v\rangle = a\langle u|v\rangle = b\langle u|v\rangle,$$

and, since $a \neq b$ by assumption, we see that this implies that $\langle u|v\rangle = 0$.

Exercise 2. Unitary Operators. An operator U is called **unitary** if $UU^\dagger = U^\dagger U = 1$. This problem shows that in an infinite dimensional Hilbert space, if an operator preserves the norm of all vectors and possesses both a left and right inverse, then it is unitary.

- a) Let U be some operator on a finite-dimensional vector space, and let $|\psi'\rangle = U|\psi\rangle$. Show that $\langle\psi'|\psi'\rangle = \langle\psi|\psi\rangle$ for all kets $|\psi\rangle$ if and only if U is unitary.
- b) Consider the Hilbert space for a particle moving in one dimension (which is infinite-dimensional). Let $|n\rangle$ be the usual harmonic oscillator eigenbasis, with $n = 0, 1, \dots$. Define an operator U by

$$U|n\rangle = |n+1\rangle.$$

Show that if $|\psi'\rangle = U|\psi\rangle$, then $\langle\psi'|\psi'\rangle = \langle\psi|\psi\rangle$. Then show that

$$U^\dagger U = \mathbb{1},$$

so that U has a left inverse, and it is $U^\dagger$. Does U have a right inverse? Is U unitary?

- a) If U is unitary, then $UU^\dagger = \mathbb{1}$, so $\langle\psi'|\psi'\rangle = \langle\psi|U^\dagger U|\psi\rangle = \langle\psi|\psi\rangle$ for all ψ . On the other hand, if U preserves the norm of all vectors, then you can use the polarization identity to win: An arbitrary inner product satisfies

$$\langle n|m\rangle = \frac{1}{4}(\langle n+m|n+m\rangle - \langle n-m|n-m\rangle),$$

so we see that inner products of the form $\langle n|UU^\dagger|m\rangle$ or $\langle n|U^\dagger U|m\rangle$ can always be expressed in terms of norms.

- b) Note that the formula

$$\langle n|U^\dagger U|n\rangle = \langle n+1|n+1\rangle = 1 = \langle n|n\rangle$$

is true for any basis vector $|n\rangle$ (since they are normalized). Given an arbitrary vector $|\psi\rangle = \sum \psi_n |n\rangle$, we can use orthonormality of the $|n\rangle$ to conclude that

$$\langle\psi|U^\dagger U|\psi\rangle = \sum_{n,m} \psi_n^* \psi_m \langle n|U^\dagger U|m\rangle = \sum_{n,m} \psi_n^* \psi_m \langle n+1|m+1\rangle = \sum |\psi_n|^2 = \langle\psi|\psi\rangle.$$

Now, we want to show that $U^\dagger U = 1$. This follows from the definition of U . Indeed, the previous calculation shows that the diagonal matrix elements of $U^\dagger U$ are the same as the identity matrix, and we just need to check that the off-diagonal matrix elements vanish. Let's check:

$$\langle n|U^\dagger U|m\rangle = 0$$

if $n \neq m$ by the definition of U . Similarly, we could do the same for $UU^\dagger$: Consider

$$\langle m|U^\dagger|n\rangle.$$

By letting $U^\dagger$ act on $\langle m|$, we see that this is the same as

$$\langle m+1|n\rangle = \delta_{m+1,n},$$

so that if we write $U^\dagger|n\rangle$ in terms of the basis elements $|m\rangle$, we immediately see that

$$U^\dagger|n\rangle = |n+1\rangle,$$

just as for U ! This means $U^\dagger = U$, but we see that $UU^\dagger|n\rangle = |n+2\rangle \neq |n\rangle$. Thus, $UU^\dagger \neq 1$. This means U is *not* unitary!

Exercise 3. *Some More Unitary Operators.* Let H be a hermitian operator and $U = e^{iH}$.

- a) Show that U is unitary. (Notice the analogy with scalars: If θ is real, then $e^{i\theta}$ has unit modulus.)
- b) If U_1, U_2 are unitary operators, show that $U_1 U_2$ is unitary. If $U_1 U_2$ is unitary, does that imply that U_1 and U_2 are?
- c) **Challenge.** Show that $\det(U) = e^{i \operatorname{tr}(H)}$.
- a) Since H is hermitian, $(e^{iH})^\dagger = e^{-iH}$. Since H commutes with itself, this implies that $e^{iH}(e^{iH})^\dagger = (e^{iH})^\dagger e^{iH} = 1$.
- b) We check

$$(U_1 U_2)^\dagger = U_2^\dagger U_1^\dagger,$$

so

$$(U_1 U_2)^\dagger U_1 U_2 = 1.$$

Similarly,

$$U_1 U_2 = (U_1 U_2)^\dagger = 1.$$

- c) We diagonalize $H = V D V^\dagger$ with V unitary. Then

$$U = V e^{iD} V^\dagger,$$

and

$$\det U = \det V \det(e^{iD}) \det(V^\dagger) = \det(e^{iD}).$$

But

$$e^{iD} = \operatorname{diag}(e^{id_1}, \dots, e^{id_n}),$$

where $D = \operatorname{diag}(d_1, \dots, d_n)$. This gives the result.

Exercise 4. *More Fun with Pauli Matrices.* Consider the 2×2 matrices,

$$\mathbb{1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The matrices $\sigma_x, \sigma_y, \sigma_z$ are called the **Pauli matrices**; we will also denote them $\sigma_1, \sigma_2, \sigma_3$.

- a) Show that the result from homework 5,

$$\sigma_i \sigma_j = \delta_{ij} + i \varepsilon_{ijk} \sigma_k, \tag{1}$$

implies that

$$[\sigma_i, \sigma_j] = 2i \varepsilon_{ijk} \sigma_k$$

and

$$\{\sigma_i, \sigma_j\} = \sigma_i \sigma_j + \sigma_j \sigma_i = 2\delta_{ij}.$$

- b) What we mean by a **vector operator** is a vector of operators; for example, $\mathbf{A} = (A_x, A_y, A_z)$, where A_x, A_y , and A_z are operators. Two vector operators are said to commute if all of their components commute. Show that, in general, $\mathbf{A} \cdot \mathbf{B} \neq \mathbf{B} \cdot \mathbf{A}$ and $\mathbf{A} \times \mathbf{B} \neq -\mathbf{B} \times \mathbf{A}$.

- c) Let $\hat{n}$ be an arbitrary unit vector and θ an arbitrary angle. Show that

$$\exp(-i\theta \boldsymbol{\sigma} \cdot \hat{n}) = \mathbb{1} \cos \theta - i(\boldsymbol{\sigma} \cdot \hat{n}) \sin \theta.$$

We often drop the $\mathbb{1}$, it being understood that a scalar like $\cos \theta$ is multiplied by the identity matrix.

Hint: The power series for sine and cosine will be useful here.

- d) Show that if A and B are square matrices, then

$$\text{tr}(AB) = \text{tr}(BA).$$

- e) Show that $\text{tr}(\sigma_i) = 0$. Then, use Equation 1 to find a simple expression for $\text{tr}(\sigma_i \sigma_j)$.

- f) The set of four 2×2 matrices $\{\mathbb{1}, \boldsymbol{\sigma}\}$ forms a basis in the space of 2×2 matrices; that is, an arbitrary 2×2 matrix can be expressed as a linear combination of these four matrices. Thus, if M is some 2×2 matrix, we can always write

$$M = a\mathbb{1} + \mathbf{b} \cdot \boldsymbol{\sigma},$$

which follows from the fact that $\mathbb{1}$ and the Pauli matrices are linearly independent, as can be seen by just staring at the four matrices. Find nice expressions for the expansion coefficients a and $\mathbf{b} = (b_1, b_2, b_3)$. Do this by taking traces or by multiplying by a Pauli matrix and then taking traces. Show that M is hermitian if and only if a and b_i are real.

- g) Express the following as linear combinations of the Pauli matrices and $\mathbb{1}$.

i) $(\mathbb{1} + i\sigma_x)^{1/2}$

Hint: Relate it to a certain half-rotation.

ii) $(2\mathbb{1} + \sigma_x)^{-1/2}$.

iii) σ_x^{-1} .

- h) Suppose that M is only nonzero in the (r, s) index, where it has a 1. That is, suppose we can write

$$M_{ij} = \delta_{ir} \delta_{js}.$$

Use this to find a nice expression for $(\sigma_m)_{ij}(\sigma_m)_{kl}$, where we again use the convention that the m index is summed over, in terms of Kronecker deltas.

- a) We compute

$$\begin{aligned} [\sigma_i, \sigma_j] &= \delta_{ij} + i\varepsilon_{ijk}\sigma_k - \delta_{ji} - i\varepsilon_{jik}\sigma_k \\ &= 2i\varepsilon_{ijk}\sigma_k, \end{aligned}$$

since $\delta_{ij} = \delta_{ji}$ and $\varepsilon_{ijk} = -\varepsilon_{jik}$. For the anticommutator, we compute

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij}$$

for the same reason as the previous identity.

b) For $\mathbf{A} \cdot \mathbf{B}$, let $\mathbf{A} = \mathbf{x}$ and $\mathbf{B} = \mathbf{p}$. Then $\mathbf{A} \cdot \mathbf{B} = x_i p_i \neq p_i x_i$ since x_i and p_i don't commute. For the cross product, we can't use the same thing, because it actually turns out that $\mathbf{x} \times \mathbf{p} = \mathbf{p} \times \mathbf{x}$ (since x_i and p_j commute when $i \neq j$). But we could define a silly version of $\mathbf{B}$ where instead of $\mathbf{B} = \mathbf{p}$ we take $\mathbf{B} = (x_2, x_1, x_3)$, which will cause same-index x 's and p 's next to each other when we take $\mathbf{p} \times \mathbf{B}$.

c) We expand the exponential

$$\exp(-i\theta \boldsymbol{\sigma} \cdot \hat{n}) = \sum \frac{(-i\theta)^m}{m!} (\boldsymbol{\sigma} \cdot \hat{n})^m.$$

Write $\boldsymbol{\sigma} \cdot \hat{n} = n_i \sigma_i$, so that

$$(\boldsymbol{\sigma} \cdot \hat{n})^2 = (n_i \sigma_i)^2 = n_i n_j \sigma_i \sigma_j = n_i n_j (\delta_{ij} + i \varepsilon_{ijk} \sigma_k) = n_i n_i = 1,$$

since the sum $n_i n_j \varepsilon_{ijk}$ vanishes by antisymmetry of ε . Thus,

$$\exp(-i\theta \boldsymbol{\sigma} \cdot \hat{n}) = \sum \frac{(-i\theta)^{2j}}{(2j)!} + \sum \frac{(-i\theta)^{2k+1}}{(2k+1)!} \boldsymbol{\sigma} \cdot \hat{n}.$$

Now, notice that

$$(-i)^{2j} = (-1)^j$$

and

$$(-i)^{2k+1} = -i(-1)^k,$$

so that we immediately recognize the power series for cosine and sine, respectively.

d) This part was done in discussion worksheet 1 from week 5.

e) The first part follows since σ_x and σ_y have zeros on the diagonal, while σ_z had $\text{tr}(\sigma_z) = 1 - 1 = 0$. We compute

$$\text{tr}(\sigma_i \sigma_j) = \text{tr}(\delta_{ij} + i \varepsilon_{ijk} \sigma_k) = 2\delta_{ij},$$

since the trace of the identity matrix is 2.

f) We immediately see that

$$\text{tr}(M) = 2a,$$

so that we can compute $a = \frac{1}{2} \text{tr}(M)$. On the other hand,

$$\sigma_i M = a\sigma_i + b_j \sigma_i \sigma_j,$$

so

$$\text{tr}(\sigma_i M) = 2b_j \delta_{ij} = 2b_i.$$

So we see that

$$b_i = \frac{1}{2} \text{tr}(\sigma_i M).$$

Since the Pauli matrices and $\mathbb{1}$ are hermitian, we see that

$$M^\dagger = a^* \mathbb{1} + \mathbf{b}^* \cdot \boldsymbol{\sigma},$$

which will be equal to $M^\dagger$ if and only if $a^* = a$ and $b_i^* = b_i$.

g) i) Following the hint, notice that

$$\sqrt{2}e^{-i\frac{\pi}{4}\sigma_x} = \mathbb{1} - i\sigma_x,$$

using the formula from (c). Thus, its (principal) square root is

$$2^{1/4}e^{i\frac{\pi}{8}\sigma_x} = 2^{1/4}(\cos(\pi/8) - i\sin(\pi/8)\sigma_x) = (\mathbb{1} + i\sigma_x)^{1/2}.$$

ii) For this one, we should observe that since

$$\sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i},$$

then if $\theta = -i\alpha$,

$$\sin(\theta) = \frac{1}{i} \sinh(\alpha).$$

Similarly,

$$\cos(\theta) = \cosh(\alpha)$$

So that

$$e^{-i(-i\alpha)\sigma_x} = \cosh(\alpha) + \sinh(\alpha)\sigma_x.$$

Now, we want to identify this with $\mathbb{1} + \frac{1}{2}\sigma_x$ (we'll add in the factor of $2^{-1/2}$ at the end). This leads us to the equation

$$\cosh(\alpha) = 2 \sinh(\alpha),$$

by setting the above equation equal to $\mathbb{1} + \frac{1}{2}\sigma_x$. Solving this for α , we find

$$\alpha = \frac{1}{2} \ln(3).$$

Now, we can take the reciprocal principal square root of our result to obtain (note that because $\sigma_x = \sigma_x^{-1}$, $(e^{\frac{1}{2}\alpha\sigma_x})^{-1} = e^{-\frac{1}{2}\alpha\sigma_x}$)

$$e^{\frac{1}{2}\alpha\sigma_x} = \exp\left(\frac{1}{4} \ln(3)\sigma_x\right) = \left(\mathbb{1} + \frac{1}{2}\sigma_x\right)^{-1/2}.$$

Finally, we can add in the factor of $2^{-1/2}$ to find

$$(2\mathbb{1} + \sigma_x)^{-1/2} = 2^{-1/2} \left[\cosh\left(\frac{1}{4} \ln(3)\right) + \sinh\left(\frac{1}{4} \ln(3)\right) \sigma_x \right].$$

Alternative Solution: We can guess that since no σ_y or σ_z appear in $(2\mathbb{1} + \sigma_x)^{-1/2}$, then we can write it as $a\mathbb{1} + b\sigma_x$. Squaring this should give us the inverse to $2\mathbb{1} + \sigma_x$, so we can deduce that

$$(a^2 + b^2 + 2ab\sigma_x)(2\mathbb{1} + \sigma_x) = \mathbb{1}.$$

We can then solve this equation for a, b to obtain the result above.

iii) $\sigma_x^{-1} = \sigma_x$.

h) By part (f), we can write

$$M_{ij} = \frac{1}{2} \text{tr}(M_{ij}) + \frac{1}{2} \text{tr}(\sigma_m M_{ij}) \sigma_m.$$

We compute

$$\text{tr}(M_{ij}) = \delta_{ij}$$

and

$$\text{tr}(\sigma_m M_{ij}) = (\sigma_m)_{kl} (M_{ij})_{kl} = (\sigma_m)_{kl} \delta_{ik} \delta_{jl} = (\sigma_m)_{ij}.$$

Thus,

$$M_{ij} = \frac{1}{2} \delta_{ij} + \frac{1}{2} (\sigma_m)_{ij} \sigma_m.$$

But we also know

$$(M_{ij})_{kl} = \delta_{ik} \delta_{jl}.$$

Taking kl components of both sides of the previous equation and setting them equal, we see that

$$\delta_{ik} \delta_{jl} = \frac{1}{2} \delta_{ij} \delta_{kl} + \frac{1}{2} (\sigma_m)_{ij} (\sigma_m)_{kl},$$

so that

$$(\sigma_m)_{ij} (\sigma_m)_{kl} = 2\delta_{ij} \delta_{kl} - \delta_{ik} \delta_{jl}.$$

Exercise 5. Even More Fun with Pauli Matrices. In this exercise, you will give an explanation of the following fact from Week 5 Worksheet 2. Let $\hat{n}$ be a unit vector. By Exercise 2(a) of the same worksheet, $\hat{z}$ can be rotated into $\hat{n}$ using just two angles. Let R be the (3×3) -matrix which rotates $\hat{z}$ into $\hat{n}$, i.e. $R\hat{z} = \hat{n}$. Now, let $U = e^{-\frac{i}{2}\alpha\sigma_z} e^{-\frac{i}{2}\beta\sigma_y}$, then

$$U^\dagger \boldsymbol{\sigma} U = R\boldsymbol{\sigma},$$

where on the RHS we understand R as rotating the components of the *vector* $\boldsymbol{\sigma}$, just as it would rotate the components of a vector $\mathbf{a}$ whose entries were scalars. We can also understand this same rotation matrix U as $U = \exp\left(-\frac{i}{2}\theta\hat{n} \cdot \boldsymbol{\sigma}\right)$.

a) Prove the identities

$$[\hat{n} \cdot \sigma, \sigma] = -2i \hat{n} \times \sigma, \quad (\hat{n} \cdot \sigma) \sigma (\hat{n} \cdot \sigma) = 2\hat{n}(\hat{n} \cdot \sigma) - \sigma.$$

b) Use (a) and Exercise 4(c) to show that for $U = \exp\left(-\frac{i}{2}\theta \hat{n} \cdot \sigma\right)$,

$$U^\dagger \sigma U = \cos(\theta) \sigma + (1 - \cos \theta) \hat{n}(\hat{n} \cdot \sigma) + \sin(\theta) \hat{n} \times \sigma.$$

c) **Challenge.** Give a physical explanation for why the result of (c) is reasonable for rotations, and why it should be the same as $R\sigma$.

a) We compute

$$[\hat{n} \cdot \sigma, \sigma] = n_i [\sigma_i, \sigma_j \hat{e}_j] = 2i n_i \varepsilon_{ijk} \hat{e}_j \sigma_k = -2i n_i \varepsilon_{ikj} \sigma_k \hat{e}_j = -2i \hat{n} \times \sigma.$$

Similarly,

$$\begin{aligned} \hat{n} \cdot \sigma \sigma \hat{n} \cdot \sigma &= n_i n_j \sigma_i \sigma_k \sigma_j \hat{e}_k \\ &= n_i n_j (\delta_{ik} + i \varepsilon_{ikl} \sigma_l) \sigma_j \hat{e}_k \\ &= n_i n_j \hat{e}_i \sigma_j + i n_i n_j \varepsilon_{ikl} (\delta_{lj} + i \varepsilon_{ljm} \sigma_m) \hat{e}_k. \end{aligned}$$

Now, the first term in the above expression is just $\hat{n}(\hat{n} \cdot \sigma)$. The first term in parentheses is 0, because the Kronecker delta turns it into $n_i n_j \varepsilon_{ikj} = 0$ by antisymmetry of ε and symmetry of $n_i n_j$ in $i \leftrightarrow j$. On the other hand, the last term can be evaluated by the identity

$$\varepsilon_{ijk} \varepsilon_{nmk} = \delta_{in} \delta_{jm} - \delta_{im} \delta_{jn},$$

which gives

$$\begin{aligned} \hat{n} \cdot \sigma \sigma \hat{n} \cdot \sigma &= \hat{n}(\hat{n} \cdot \sigma) - n_i n_j \sigma_m \hat{e}_k (\delta_{ij} \delta_{km} - \delta_{im} \delta_{kj}) \\ &= 2\hat{n}(\hat{n} \cdot \sigma) - \sigma \end{aligned}$$

as desired.

b) Write $U = e^{-i\frac{\theta}{2}\hat{n} \cdot \sigma}$ as representing a rotation by angle θ about the axis $\hat{n}$. We can then, using the identities above, compute

$$\begin{aligned} U^\dagger \sigma U &= \cos^2\left(\frac{\theta}{2}\right) \sigma + i \cos\left(\frac{\theta}{2}\right) \sin\left(\frac{\theta}{2}\right) [\hat{n} \cdot \sigma, \sigma] + \sin^2\left(\frac{\theta}{2}\right) \hat{n} \cdot \sigma \sigma \hat{n} \cdot \sigma \\ &= [\cos^2(\theta/2) - \sin^2(\theta/2)] \sigma - 2 \cos(\theta/2) \sin(\theta/2) \hat{n} \times \sigma + 2 \sin^2(\theta/2) \hat{n}(\hat{n} \cdot \sigma). \end{aligned}$$

Then, we can use the trigonometric half-angle identities to get our answer in the required form.

c) If σ was an ordinary 3-vector, then we can interpret the result for $U^\dagger \sigma U$ as follows. The rotation keeps the component parallel to $\hat{n}$ invariant, and it rotates the component perpendicular to $\hat{n}$ by an angle θ in the perpendicular plane.

Exercise 6. Ignore the fact that the hydrogen atom is a three-dimensional problem, and imagine that

$$H = \frac{p^2}{2m} - \frac{e^2}{r}$$

corresponds to a one-dimensional problem, where $p^2 = p_x^2 + p_y^2 + p_z^2$ and $r^2 = x^2 + y^2 + z^2$. Assuming that

$$\Delta p \Delta r \geq \hbar/2,$$

estimate the ground state energy. (The actual ground state energy is about -13.6 eV).

Hint: To make your calculation easier, recall that $mc^2 = 511$ keV is the electron rest energy. You can use that the Bohr radius is

$$\begin{aligned} a_0 &= \frac{\hbar^2}{me^2} \\ &\approx 137 \cdot \frac{\hbar}{mc}. \end{aligned}$$

Assuming $\Delta p \sim \hbar/\Delta r$, we can plug in to the hamiltonian to find

$$H = \frac{\hbar^2}{2mr^2} - \frac{e^2}{r}.$$

The ground state is a minimum for this function, so we can take a derivative with respect to r and set it equal to 0 to find

$$0 = -2\frac{\hbar^2}{2mr^3} + e^2/r^2.$$

Solving this for r , we get

$$r_{\min} = \frac{\hbar^2}{me^2},$$

which is the Bohr radius. Using the hint, we can plug it back into H to find

$$H_{\min} = E_{\min} = 13.6 \text{ eV} - 27.2 \text{ eV} = -13.6 \text{ eV}$$

which is the correct answer!!!